

Trainings ▾

General Information

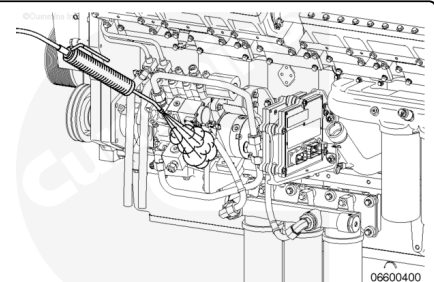
Revised diagnostic hardware was released on K38 and K50 engines in October, 1993 (engine serial number first 33125767). This included hardware installed along with a revised fuel block designed for usage with the single cylinder diagnostic kit. The original diagnostic tool was released in March, 1993, (Part Number 3824212). The kit was updated to include four needle valves designed to work in the revised fuel block.

Engines equipped with the revised fuel blocks and related hardware reduce the time involved in isolating single cylinder problems. Studies indicated a time saving of 45 minutes when using this hardware.

Preparatory Steps

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Some solvents are flammable and toxic. Read the manufacturer's instructions before use.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause bodily injury.

- Thoroughly clean debris from all fittings and components before removal. Make sure that debris, water, steam, or cleaning solution does **not** get inside the fuel system.

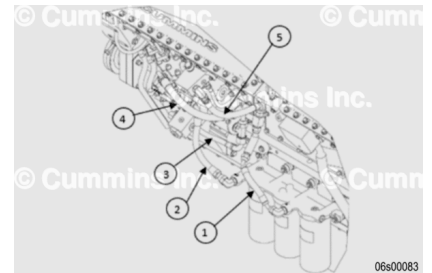
- Use compressed air to dry fittings and components.

Remove

QSK38 and QSK50 Engines with Gerotor Fuel Pump Drain Line

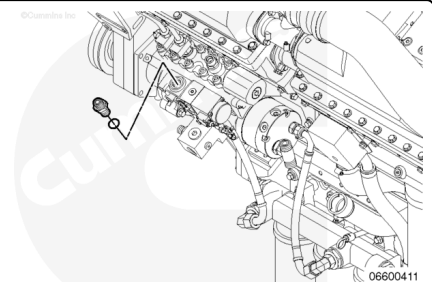
Remove the following and cover the fittings with a plastic bag, or place the items in a plastic bag.

1. Outlet of the gerotor to inlet of stage two filter
2. Outlet of stage two filter to the inlet of the fuel pump pressurizing assembly
3. Fuel supply to gerotor
4. Fuel bypass to gerotor
5. Fuel drain line to gerotor.



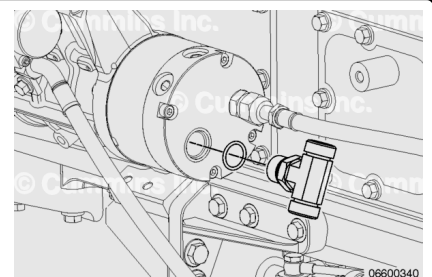
Remove the following and cover the fittings with a plastic bag, or place the items inside a plastic bag.

Remove the union from the fuel pump pressurizing assembly and discard the o-ring.



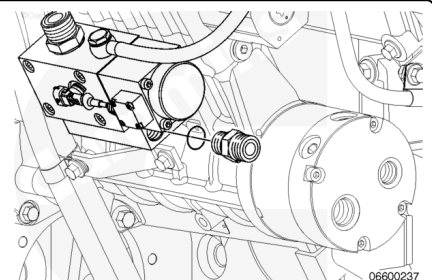
Remove the following and cover the fittings with a plastic bag, or place the items inside a plastic bag.

Remove the tee fitting from the gerotor fuel pump. Discard the o-rings.



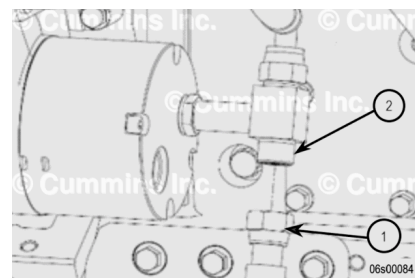
Remove the following and cover the fittings with a plastic bag, or place the items in a plastic bag.

Remove the union from the inlet of the fuel pump actuator. Discard the o-rings.



QSK38 and QSK50 Engines with Gerotor Fuel Pump Drain Line:

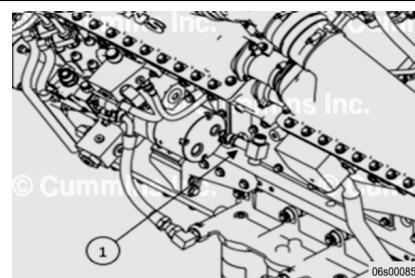
Remove the 45-degree extension (1) from the lower T Piece male union (2) and cover the fitting with a plastic bag or place the part in a plastic bag.



QSK38 and QSK50 Engines with Gerotor Fuel Pump Drain Line:

Remove extended T-Piece fitting (1) and cover the fitting with a plastic bag or place the items in a plastic bag.

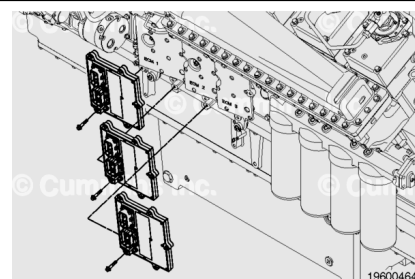
Discard the exterior o-ring.



QSK50 Engines with Steel Braided Lines:

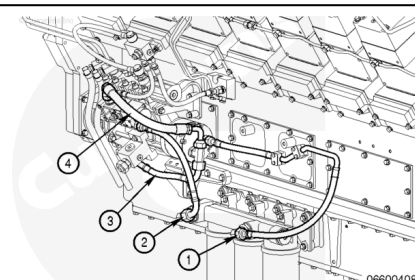
If the outlet of the gerotor to the inlet of the stage two filter line needs to be removed, the ECMs and ECM cooling plate may need to be removed.

- Use the following procedure in the QSK50 and QSK60 Modular Common Rail System Troubleshooting and Repair Manual, Bulletin 4021533 to remove the ECMs. Refer to Procedure 019-031 in Section 19. (</qs3/pubsys2/xml/en/procedures/105/105-019-031.html>)
- Remove the ECM cooling plate. Refer to Procedure 008-128 in Section 8. (</qs3/pubsys2/xml/en/procedures/28/28-008-128.html>)



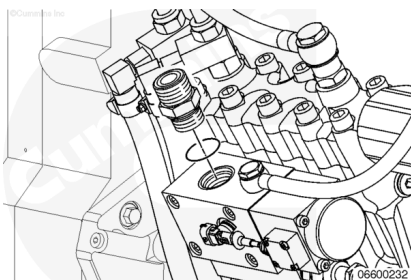
Remove the following and cover the fittings with a plastic bag, or place the items in a plastic bag.

1. Outlet of the gerotor to the inlet of the stage two filter
2. Outlet of the stage two filter to the inlet of the fuel pump pressurizing assembly
3. Fuel supply to gerotor
4. Fuel bypass to gerotor.



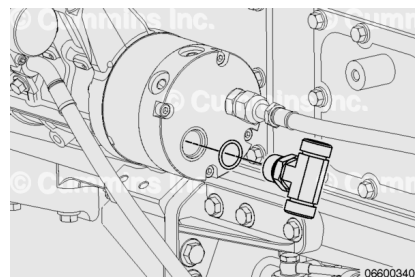
Remove the following and cover the fittings with a plastic bag, or place the items inside a plastic bag.

Remove the union from the fuel pump pressurizing assembly and discard the o-ring.



Remove the following and cover the fittings with a plastic bag, or place the items inside a plastic bag.

Remove the tee fitting from the gerotor fuel pump. Discard the o-rings.

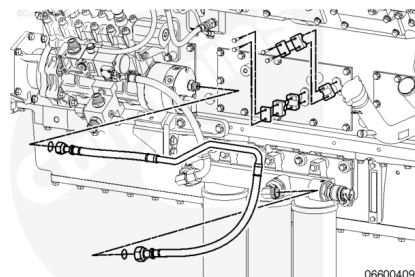


Remove the following and cover the fittings with a plastic bag, or place the items in a plastic bag.

Remove the four capscrews and clamps securing the fuel line to the block.

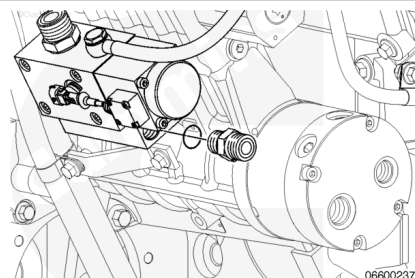
Remove the fuel supply line from the gerotor pump output to the inlet of the stage two fuel filter inlet.

Discard the o-rings.



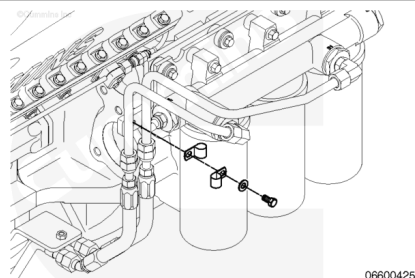
Remove the following and cover the fittings with a plastic bag, or place the items in a plastic bag.

Remove the union from the inlet of the fuel pump actuator. Discard the o-rings.



Marine Applications

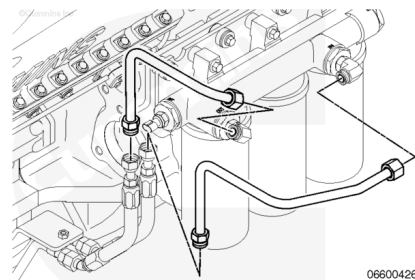
Remove the capscrews and clamps securing the fuel line to the mounting plates at the stage two filter head.



Remove the following and cover the fittings with a plastic bag, or place the items in a plastic bag.

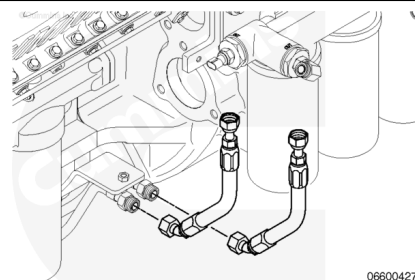
Remove the short section of the fuel supply lines to the inlet and outlet of the stage two filter head.

Discard the o-rings.

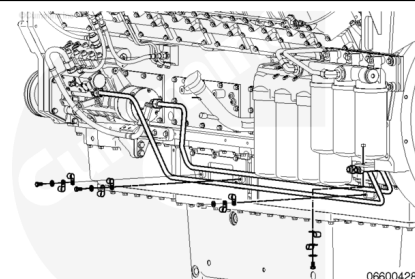


Remove the following and cover the fittings with a plastic bag, or place the items in a plastic bag.

Remove the two flexible fuel lines.



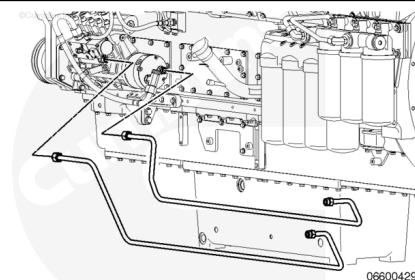
Remove the capscrews and clamps securing the fuel line to the mounting plates.



Remove the following and cover the fittings with a plastic bag, or place the items in a plastic bag.

Disconnect the fuel supply lines at the fuel pump and remove the fuel supply lines to the remote mounted stage two filter.

Discard the o-rings.



Clean and Inspect for Reuse

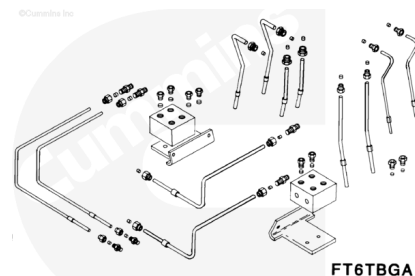
with Mechanically Actuated Injector

⚠ WARNING ⚠

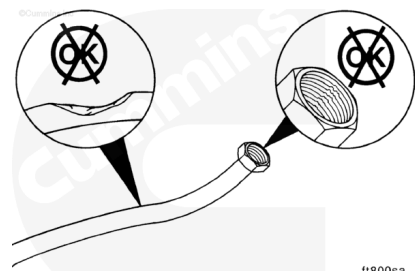
When using solvents, acids or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to avoid personal injury.

Remove and discard the ferrule seals.

Use solvent to flush the tubing and the fuel block connectors.



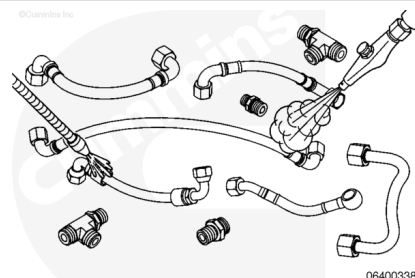
Inspect the fuel tubes for cracks or locations where the tubing is crushed. Do **not** attempt to repair fuel tubing that is damaged.



with Electronically Actuated Injector

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Some solvents are flammable and toxic. Read the manufacturer's instructions before use.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause bodily injury.

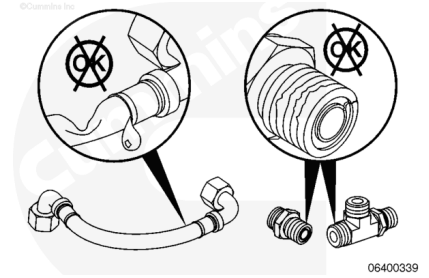
Use solvent to clean the fuel supply lines, tee fittings, and unions.

Use compressed air to dry the lines and fittings.

Inspect the fuel supply lines for restrictions that can possibly prevent fuel from flowing freely. Replace the fuel supply lines if they are kinked or damaged, or if leakage is found.

Inspect the tee fittings and unions for damage. Replace if damaged, cracked, or stripped threads are found.

Cover the fuel line fittings with a plastic bag, or place the items inside a plastic bag after cleaning and inspection.



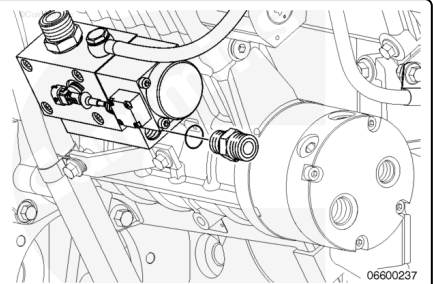
Install

with Electronically Actuated Injector

Remove the union from the protective plastic bag.

Use a new o-ring and install the union in the inlet of the fuel pump pressurizing assembly.

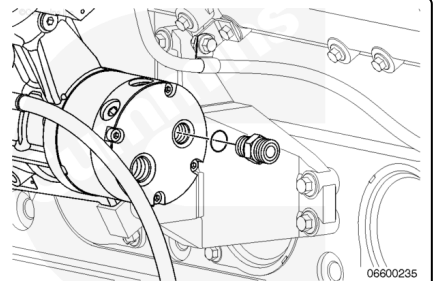
Torque Value: 100 n·m [74 ft-lb]



Remove the union from the protective plastic bag.

Install new o-rings on the union and install the union in the outlet of the gerotor fuel pump.

Torque Value: 100 n·m [74 ft-lb]



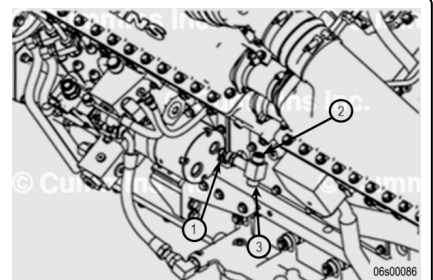
QSK38 and QSK50 Engines with Gerotor Fuel Pump Drain Line:

Remove extended T-Piece fitting from plastic bag.

Install new o-ring (1) and check original o-rings are still located at upper (2) and lower (3) outlets. Install union into the rear outlet of the gerotor pump with arrow on valve pointing vertically upwards.

Install the T-Piece fitting until the backup washer contacts the port face. Use two wrenches to hold the T-Piece fitting in the desired position and tighten the locknut.

Torque Value: 80 n·m [59 ft-lb]

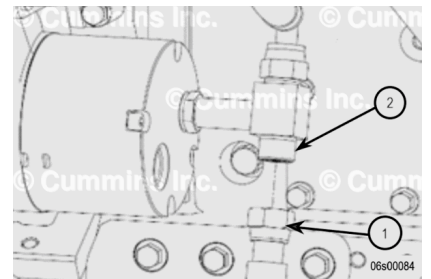


QSK38 and QSK50 Engines with Gerotor Fuel Pump Drain Line:

Remove the 45-degree extension (1) from plastic bag.

Install to the lower T-Piece male union (2) facing as shown in the image.

Torque Value: 80 n•m [59 ft-lb]



Remove the fuel line from the protective plastic bag. Install the fuel line from the outlet of the stage two filters to the inlet of the fuel pump to the pressurizing assembly.

Tighten the hose fitting (1) to the elbow of the filter head.

Tighten the hose fitting at the union (2) on the fuel pump pressurizing assembly.

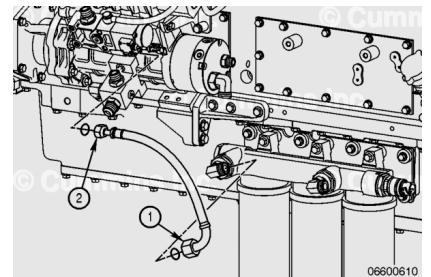
Note : Hose fitting sizes differ, depending on the hose part number.

Torque Value:

1 - 14 UNF Fittings 65 n•m [48 ft-lb]

Torque Value:

1 3/16 - 12 UNF Fittings 90 n•m [66 ft-lb]



Remove the fuel line from the protective plastic bag.

Install the fuel supply line from the gerotor fuel pump to the inlet of the stage two fuel filter.

Tighten the hose fitting (1) to the elbow of the filter head.

Tighten the hose fitting at the union (2) on the gerotor fuel pump.

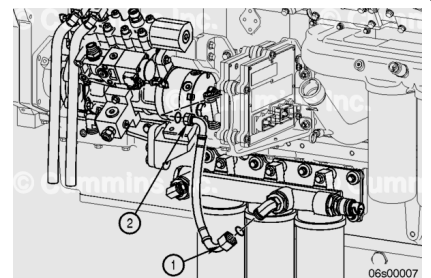
Note : Hose fitting sizes differ, depending on hose part number.

Torque Value:

1 - 14 UNF Fittings 65 n•m [48 ft-lb]

Torque Value:

1 3/16 - 12 UNF Fittings 90 n•m [66 ft-lb]

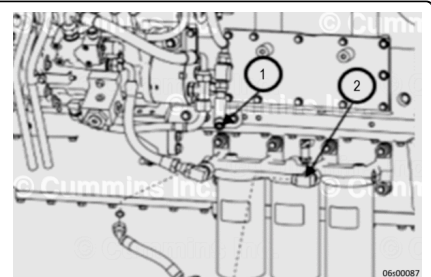


QSK38 and QSK50 Engines with Gerotor Fuel Pump Drain Line:

Remove the fuel line from the plastic bag.

Install the fuel line between the gerotor elbow (1) and the inlet of the stage two filter (2).

Torque Value:

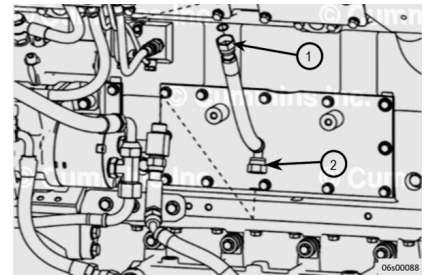


1. 80 n•m [59 ft-lb]
2. 80 n•m [59 ft-lb]

QSK38 and QSK50 Engines with Gerotor Fuel Pump Drain Line:

Remove the fuel drain line from the protective plastic bag.
Replace the o-ring.

Install the line between the fuel manifold drain elbow (1), and
the upper connect of the gerotor rear t-piece (2).



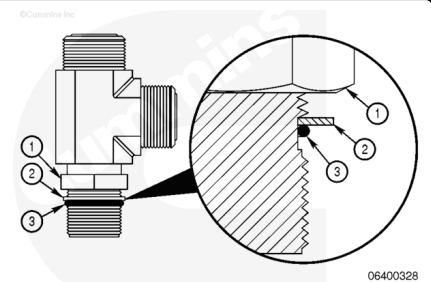
Torque Value:

1. 80 n•m [59 ft-lb]
2. 80 n•m [59 ft-lb]

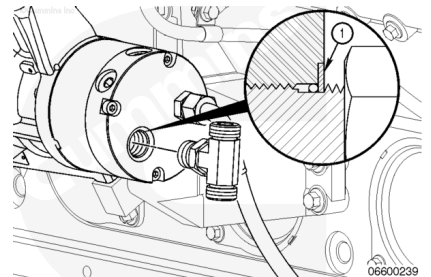
Remove the tee fitting from the protective plastic bag.

Install the tee fitting in the port on the back of the fuel pump by
first positioning the locknut.

1. Locknut
2. Washer
3. O-ring.

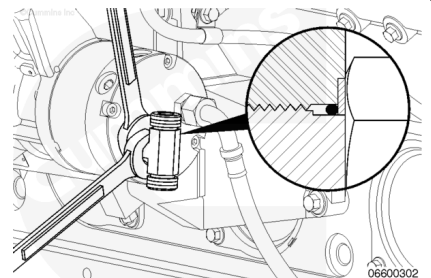


Install the tee fitting in the fuel pump port until the backup
washer (1) contacts the port face.



Use two wrenches to hold the tee fitting in the desired position
and tighten the locknut.

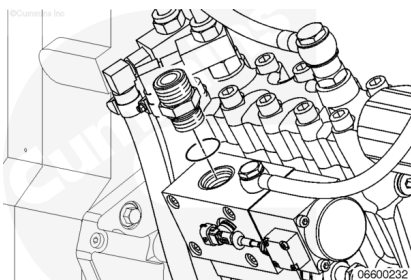
Torque Value: 100 n•m [74 ft-lb]



Remove the union from the protective plastic bag.

Install new o-rings in the union and install it in the port for the
bypass from the fuel pump inlet actuator.

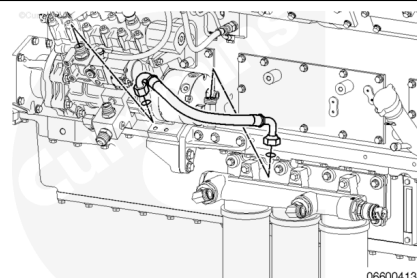
Torque Value: 170 n•m [125 ft-lb]



Remove the fuel supply line from the protective plastic bag.

Install the fuel bypass line on the inlet actuator bypass union to the tee fitting on the inlet of the gerotor fuel pump.

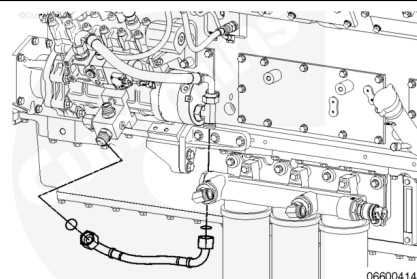
Torque Value: 91 n•m [67 ft-lb]



Remove the fuel supply line from the protective plastic bag.

Install the fuel supply line from the fuel manifold to the tee fitting on the inlet of the gerotor fuel pump.

Torque Value: 91 n•m [67 ft-lb]



QSK50 Engines with Steel Braided Lines

Install the clamps (3) and four capscrews (4) to secure the fuel line to the cylinder block.

Torque Value:

Fuel Supply Clamps 9 n•m [80 in-lb]

Remove the fuel line from the protective plastic bag.

Install the fuel line from the gerotor fuel pump to the inlet of the stage two fuel filter.

Tighten the hose fitting (1) to the elbow of the filter head.

Tighten the hose fitting at the union (2) on the gerotor fuel pump.

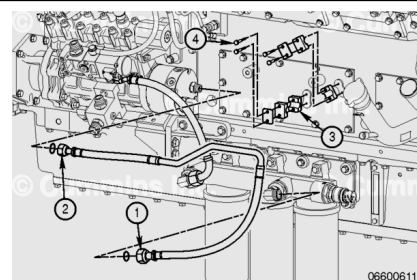
Note : Hose fitting sizes differ, depending on the hose part number.

Torque Value:

1 - 14 UNF Fittings 65 n•m [48 ft-lb]

Torque Value:

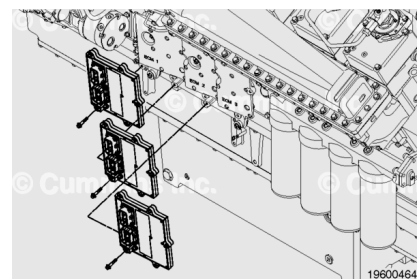
1 3/16 - 12 UNF Fittings 90 n•m [66 ft-lb]



Install the ECM cooling plate. Refer to Procedure 008-128 in Section 8. (</qs3/pubsys2/xml/en/procedures/28/28-008-128.html>)

Use the following procedure in the QSK50 and QSK60 Modular Common Rail System, Troubleshooting and Repair Manual, Bulletin 4021533, to install the ECMs. Refer to Procedure 019-031 in Section 19.

(</qs3/pubsys2/xml/en/procedures/105/105-019-031.html>)

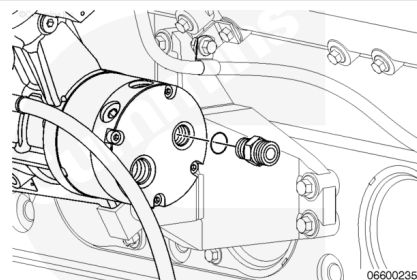


Marine Applications

Remove the union from the protective plastic bag.

Install new o-rings on the union and install the union in the outlet of the gerotor fuel pump.

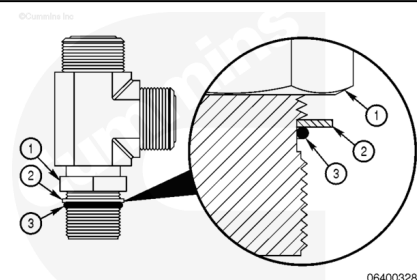
Torque Value: 100 n•m [74 ft-lb]



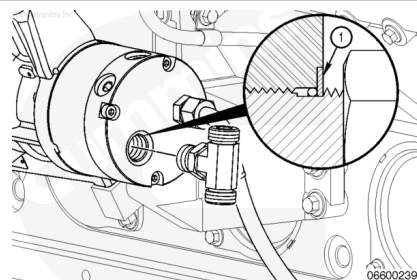
Remove the tee fitting from the protective plastic bag.

Install the tee fitting in the port on the back of the fuel pump by first positioning the locknut.

1. Locknut
2. Washer
3. O-ring.

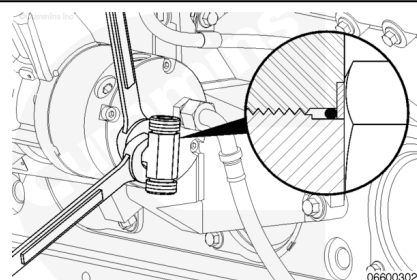


Install the tee fitting in the fuel pump port until the backup washer (1) contacts the port face.



Use two wrenches to hold the tee fitting in the desired position and tighten the locknut.

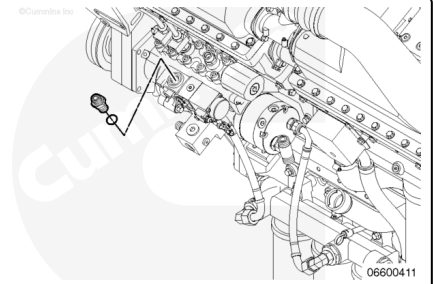
Torque Value: 100 n•m [74 ft-lb]



Remove the union from the protective plastic bag.

Install new o-rings in the union and install it in the port for the bypass from the fuel pump inlet actuator.

Torque Value: 170 n•m [125 ft-lb]

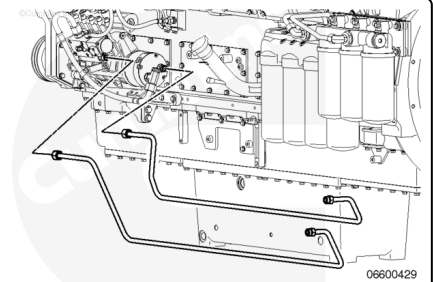


Remove the fuel supply line from the protective plastic bag.

Install the fuel supply lines that go between the fuel pump and the remote mounted stage two fuel filters.

Tighten the fuel lines at the pump.

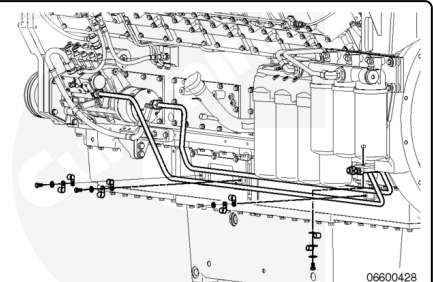
Torque Value: 65 n•m [48 ft-lb]



Install the clamps and capscrews securing the fuel line to the mounting plates.

Tighten the capscrews.

Torque Value: 9 n•m [80 in-lb]

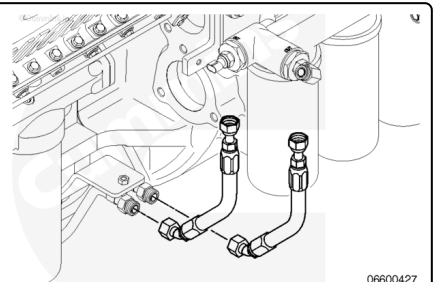


Remove the flexible fuel supply lines from the protective plastic bag.

Install the two flexible fuel lines.

Tighten the fittings.

Torque Value: 91 n•m [67 ft-lb]

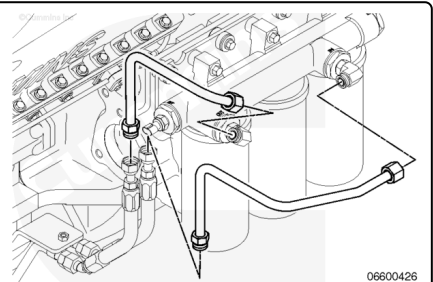


Remove the fuel supply lines from the protective plastic bag.

Install the short section of the fuel supply lines to the inlet and outlet of the stage two filter head.

Tighten the fittings.

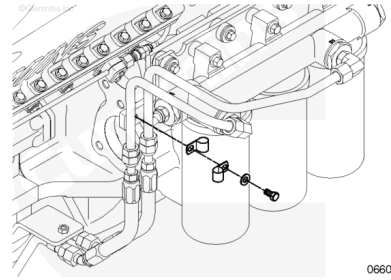
Torque Value: 91 n•m [67 ft-lb]



Install the clamps and capscrews securing the fuel line to the mounting plates at the stage 2 filter head.

Tighten the fittings.

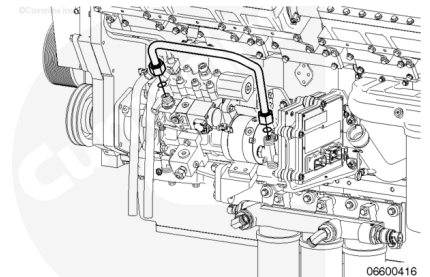
Torque Value: 9 n•m [80 in-lb]



Remove the fuel supply line from the protective plastic bag.

Install the fuel bypass line on the inlet actuator bypass union to the tee fitting on the inlet of the gerotor fuel pump.

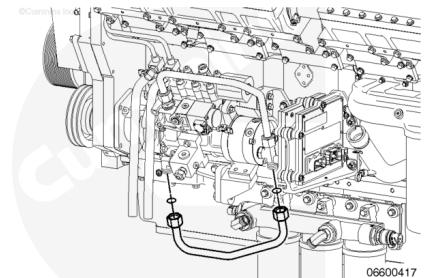
Torque Value: 91 n•m [67 ft-lb]



Remove the fuel supply line from the protective plastic bag.

Install the fuel supply line from the fuel manifold to the tee fitting on the gerotor fuel pump.

Torque Value: 91 n•m [67 ft-lb]



Finishing Steps

- Operate the engine and check for leaks. If leaks are found, repair and test again.



